

Volume Vii Reference

Contents

0.1	Category Theory	3
0.2	Languages and Structures	4
0.3	Proof Theory	7
0.4	Type Theory	8
0.5	Lambda Terms	9

Category Theory

0.1 Category Theory

Proofs

Capstone

Model Theory

0.2 Languages and Structures

First-Order Languages

Definition 0.2.1 (First-Order Language $\mathcal{L}$). A **first-order language** $\mathcal{L}$ consists of three (possibly empty) sets of non-logical symbols:

- a set $\mathcal{R}$ of **relation symbols** (also called *predicate symbols*), each assigned a fixed arity $n \geq 1$;
- a set $\mathcal{F}$ of **function symbols**, each assigned a fixed **arity** $n \geq 0$ (0-ary function symbols are *constant symbols*);
- a set $\mathcal{C}$ of **constant symbols** (equivalently, 0-ary function symbols — listed separately for emphasis).

Together with the logical symbols common to all first-order languages (variables, connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$, quantifiers $\forall, \exists$, equality $=$, parentheses), the triple $(\mathcal{F}, \mathcal{R}, \mathcal{C})$ determines $\mathcal{L}$.

L-Structures

Definition 0.2.2 ($\mathcal{L}$ -Structure). Let $\mathcal{L} = (\mathcal{F}, \mathcal{R}, \mathcal{C})$ be a first-order language. An **$\mathcal{L}$ -structure** $\mathcal{M}$ is an interpretation of the symbols of $\mathcal{L}$ on a non-empty domain.

Variable Assignments

Definition 0.2.3 (Variable Assignment). Let $\mathcal{M}$ be an $\mathcal{L}$ -structure with domain M , and let $\text{Var} = \{v_1, v_2, v_3, \dots\}$ be a fixed countable set of first-order variables.

A **variable assignment** (or **environment**) for $\mathcal{M}$ is a function $s : \text{Var} \rightarrow M$.

For a variable assignment s , a variable v_i , and an element $a \in M$, write $s[v_i \mapsto a]$ for the assignment that agrees with s on all variables except v_i , where it returns a :

$$s[v_i \mapsto a](v_j) = \begin{cases} a & \text{if } j = i, \\ s(v_j) & \text{if } j \neq i. \end{cases}$$

Term Evaluation

Definition 0.2.4 (Term Evaluation). Let $\mathcal{M}$ be an $\mathcal{L}$ -structure and $s : \text{Var} \rightarrow M$ a variable assignment. The **evaluation** $s(t) \in M$ of an $\mathcal{L}$ -term t under s is defined inductively:

1. If $t = v_i$ is a variable, then $s(t) = s(v_i)$.
2. If $t = c$ is a constant symbol, then $s(t) = c^{\mathcal{M}}$.
3. If $t = f(t_1, \dots, t_n)$ for an n -ary function symbol f , then $s(t) = f^{\mathcal{M}}(s(t_1), \dots, s(t_n))$.

Satisfaction [Stub]

Elementary Equivalence and Substructures [Stub]

Proofs

Capstone

Advanced Logic

Proof Theory

0.3 Proof Theory

Proofs

Capstone

Type Theory

0.4 Type Theory

Proofs

Capstone

Lambda Calculus

0.5 Lambda Terms

Untyped Lambda Calculus

Definition 0.5.1 (Lambda Terms). The set Λ of λ -terms is defined inductively as follows:

1. Every variable x is a λ -term.
2. If M is a λ -term and x is a variable, then

$$\lambda x. M$$

is a λ -term.

3. If M and N are λ -terms, then

$$MN$$

is a λ -term.

Proofs

Capstone